

RICHARD WOLLYCE SANTOS DE SOUZA

TECH LEAD | FULL-STACK SOFTWARE ENGINEER | AI SYSTEMS

Franca, SP, Brazil | +55 (16) 9 9159-7978 | mail@richardwollyce.com
richardwollyce.com | linkedin.com/in/richardwollyce-/ | github.com/richard-wollyce

PROFESSIONAL SUMMARY

Tech Lead and Full-Stack Software Engineer working across AI infrastructure and commerce systems. Creator of Ulpia, an open-source local-first memory layer for AI agents written in Rust, with deterministic retrieval, a first-class abstention verdict, sub-millisecond warm queries, and an MCP server; published under Apache 2.0 with its benchmarks in the same repository. Leads engineering at Casa Seth across infoproducts, Pix payments, attribution, server-side conversion tracking, financial reconciliation, and the BiblinhaPlay subscription platform. Combines architecture leadership with implementation, incident response, CI/CD, and production ownership.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, SQL; working knowledge of Python and Bash

AI & Retrieval Systems: Local-first memory architecture, deterministic retrieval, SQLite FTS5, Reciprocal Rank Fusion, abstention and confidence gating, Model Context Protocol (MCP), RAG, benchmark design

Web & Mobile: React, TanStack Start/Router/Query, Next.js, Expo, React Native, Three.js, Vite, Tailwind CSS

Backend & Data: Node.js, PostgreSQL, Supabase, Drizzle ORM, Better Auth, REST APIs, Edge Functions, RLS, RBAC

Commerce & Analytics: Pix, hosted checkout, idempotent webhooks, entitlements, revenue reconciliation, UTM attribution, server-side conversion tracking, PostHog, Mercado Pago

Platforms & Delivery: Docker, Linux, Nginx, GitHub Actions, Vercel, EAS, CI/CD, Cargo, Tauri, BaseHub

Quality: Vitest, Playwright, Maestro, TDD, automated linting, type checking, build gates, production monitoring

PROFESSIONAL EXPERIENCE

Creator & Maintainer | Ulpia (Open Source)

Apache 2.0 | ulpia.io
August 2026 - Present

- Design and build a local-first memory layer for fleets of AI agents in Rust, keeping every model out of the retrieval path so results are offline, reproducible, and explainable when wrong.
- Built a two-scorer retrieval engine over a declared-keyword index and SQLite full-text search, fused with Reciprocal Rank Fusion, and measured which scorer wins which job rather than assuming they were interchangeable.
- Made refusal a first-class verdict behind a confidence gate: on a blind, adversarially checked question set, 28 of 30 out-of-scope questions are not answered confidently by the deterministic layer alone.
- Wrote the benchmark harness alongside the product: warm route latency of 0.68 ms p50 and 1.16 ms p95 in process, and 97 percent on abstention across the 500 questions of LongMemEval-S, with every result stamped with its command, commit, machine, and date.
- Shipped an MCP server exposing four read-only tools to Claude Desktop and other MCP clients, deliberately keeping write access off the surface a model can reach.
- Maintain roughly 17,000 lines of Rust across three crates with a single runtime dependency, over 200 tests, CI on GitHub Actions, and 33 architecture decision records carrying the reasoning behind each trade.

Tech Lead & Software Engineer | Casa Seth

Brazil
April 2026 - Present

- Lead architecture and hands-on delivery across the company's infoproducts, digital commerce funnels, and the measurement systems that report what each of them earned.
- Own conversion engineering end to end: Pix checkout, server-controlled pricing, UTM attribution, first-party browser and server event tracking with deduplication, revenue reconciliation, and operational dashboards.
- Designed a generative-image pipeline with idempotent jobs, bounded concurrency, caching, persistence, and telemetry, so a failed job retries without duplicating paid work.
- Lead BiblinhaPlay, the subscription learning and entertainment platform serving roughly 500 users across a production web/PWA and an Expo/React Native client, covering video streaming, music, printables, interactive games, and gamification.
- Structured a TypeScript monorepo with pnpm, Turborepo, React, TanStack Start, shared UI and transactional-email packages, PostgreSQL/Supabase, Drizzle ORM, and automated quality gates.
- Designed server-side authentication and authorization, plan-based entitlements, hosted subscription checkout, verified idempotent webhooks, payment reconciliation, protected media streaming, and immutable content releases with atomic activation and rollback.
- Built BiblinhaCraft, a Three.js voxel experience with deterministic procedural terrain, lazy regional streaming, versioned save migration, missions, and touch-first controls.

Software Engineer | MG Laser

Franca, Brazil
November 2025 - April 2026

- Built and maintained an ERP covering inventory, sales, and daily operations across multiple teams.

- Replaced spreadsheet workflows with structured forms and automated validation, reducing manual data-entry errors.
- Improved high-volume table performance through pagination and targeted PostgreSQL RPC calls while enforcing Row-Level Security and Role-Based Access Control.
- Managed deployment, monitoring, and infrastructure on a self-managed Linux VPS; restored service after a critical production outage in under 10 minutes with no data loss.

Independent Software Engineer | Independent / Contract

Franca, Brazil
Ongoing

- Deliver full-stack web applications using TypeScript, React, Next.js, Node.js, Supabase, PostgreSQL, Vite, Tailwind CSS, and Vercel.
- Built a live-event registration system with Brazilian identity and WhatsApp validation, responsive staff workflows, and participant tracking.
- Built service-business chat and administrative workflows that turn conversations into structured budget requests and follow-up tasks.
- Own frontend, backend, database design, deployment, maintenance, and live support, using Vitest and TDD to prevent regressions.

SELECTED ENGINEERING WORK

Ulpia | Local-First AI Memory Infrastructure, Rust, Apache 2.0 | ulpia.io

- Memory layer for fleets of AI agents with no embedding model in the retrieval path, so it runs offline, returns the same answer twice, and can report that no file covers a question instead of returning the least wrong one.
- Two-scorer engine over a declared-keyword index and SQLite FTS5, fused with Reciprocal Rank Fusion, behind a confidence gate that turns abstention into a first-class verdict.
- Measured, not asserted: 28 of 30 out-of-scope questions declined by the deterministic layer on a blind adversarial set, 0.68 ms p50 warm route latency, 97 percent abstention across LongMemEval-S.
- MCP server with four read-only tools for Claude Desktop and other clients, a git-derived privacy model where an untracked file is never served, and a Tauri tray client on Windows.

Casa Seth | Infoproducts, Commerce & Conversion Systems | biblinhaplay.com

- Shared domain and UI packages across independently deployed product funnels on Supabase and Postgres Edge Functions.
- Pix checkout, server-controlled pricing, UTM attribution, browser and server event deduplication, revenue reconciliation, and operator-led physical fulfillment workflows.
- Paid-order validation, idempotent image generation, caching, concurrency control, telemetry, storage, and production-faithful benchmarking.
- BiblinhaPlay, the subscription learning and entertainment platform inside the house, serving roughly 500 users across a production web/PWA and an Expo/React Native client: video streaming, music, printables, games, entitlement-based access, HTTP Range media delivery, immutable content snapshots, and the BiblinhaCraft Three.js voxel experience.

RoadToCyberSec.com | Cybersecurity Education

- Authored a structured learning hub covering threat analysis, MFA, safe browsing, incident response, networking fundamentals, and digital evidence handling.
- Published the material through a searchable Mintlify documentation experience for technical and non-technical learners.

EDUCATION

B.Sc. in Software Engineering | Universidade de Franca | 2025 - 2029

CERTIFICATIONS

- Santander Bootcamp: Rust and AI-Integrated Application Development | June 2026
- Computational Forensics and Digital Evidence Investigation, Universidade Cruzeiro do Sul | June 2026
- LEAD1x: Exercising Leadership: Foundational Principles, HarvardX / edX | May 2026
- Networking Basics, Cisco Networking Academy | July 2025
- Introduction to Cybersecurity, Cisco Networking Academy | July 2023

LANGUAGES

Portuguese - Native | English - Advanced (C1) | Spanish - Intermediate